Simulation study design for investigating Random Forests for Variable Importance

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August 2026

1 Purpose of simulation study

The purpose of the present simulation study is to investigate the performance of various variable importance measures for random forests. An intermediate goal is to find a parameter set for random forest fitting that yield good results in a situation similar to Hindborgs study of ECG abnormalities in relation to OHCA.
Two situations that appear in practice that we would like a method to handle well are

1. strong correlations between a large group of variables where only one among them has a causal effect and
2. strong interaction effects between variables that individually have modest or no effect.

2 Simulation setting

2.1 Data generating process

We will simulate from a logistic regression model with a single binary outcome, Y and 42 binary covariates, $(X_1, \dots, X_{42} = X)$.

$$EY = logis(\theta_0 + \lambda\theta X) \tag{1}$$

The simulation engine is created from the ECG + OHCA dataset using the synthesize function of the riskRegression package. For this reason, the correlation structure of the covariates are similar to this case.
For effect size, we will assign an effect size of 1 to the first four entries of θ, and an effect size of 0.5 to the next four, with the remaining being 0: $\theta = (1, 1, 1, 1, 0.5, 0.5, 0.5, 0.5, 0, \dots, 0)$. To examine performance under different effect sizes/strength of signal, we will simulate data from different models where the effect sized are scaled by a parameter λ.

2.2 Adding interaction effects

To examine the performance of variable importance measures in the presence of interaction effects, we add an interaction term to the linear predictor. In particular, we add three interaction pairs: one pair among the variables with large first-order effect, one pair among the variables that have moderate first-order effects and one pair among the variables that have no first-order effects. In this setting the outcome model is given by

$$EY = logis(\theta_0 + \lambda(\theta X + \psi(X))) \tag{2}$$

where $\psi(X) = X_1 X_2 + X_4 X_6 + X_9 X_{10}$.

2.3 Setting with no first-order effects but presence of interaction effects

2.4 Hyperparameters

Initially, we will work with a sample size of 2000. We will investigate performance of each variable importance measure under consideration for three values for each of the following parameters for random forest fitting: mtry, nodesize, ntree. We will also investigate performance for different scales of effect size in the data generating process.
Concretely, we will use the following values:

Parameter	Low	Medium	High
mtry	3	8	16
nodesize	1	5	20
ntree	250	500	2000
λ	0.1	0.55	1

Table 1: Random forest parameter values considered in the simulation study.

3 Variable importance measures

For random forests we will consider permutation-based variable importance as well as minimal depth and average treatment effect as variable importance measures. Random forests will be fit using Brier score as loss function. For a comparison outside the random forest framework, we will use average treatment effect within a logistic regression model.

4 Performance assessment

4.1 Signal strength

We ask ourselves the question: Do the variables in our model actually predict the outcome in question? To assess this, we will, for each model, compare the prediction error of the model to a trivial model that uniformly assigns the marginal average outcome of the entire population to each individual.

4.2 Notions of scoring for rankings

Exactly how performance of variable importance measures is to be scored is unclear to me at this time. A non-parametric method of scoring would be preferable, as different notions of importance might not be directly comparable. If each variable importance measure admits a ranking (preferably allowing for ties), the accuracy of importance rankings may be evaluated against a ground truth of importance rank. The task of selecting a subset of important variables from a large set of mostly unimportant variables may be different than ranking the importance of a relatively small set of variables that is known to have some importance. These two tasks may be in tension for some methods. Perhaps it should be further clarified what kind of task we wish to score with respect to.

4.3 Ranking variable importance in the presence of interactions

When the data generating mechanism is a logistic regression with no interaction terms, the size of the parameters of the model admit a straightforward ground truth of variable importance. It is unclear to me how a one-dimensional ground truth of variable importance is to be constructed in the presence of interactions. Perhaps a computationally efficient version of Shapley values can be implemented for random forests?